MILITARY SPECIFICATION

DISTRIBUTION BOX
J-3425C/AWW

This specification is approved for use within the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, manufacture, and acceptance requirements for the J-3425C/AWW Distribution Box, herein referred to as the distribution box. The distribution box is designed to provide interface between the AN/AWW-12 or AN/AWW-9 Control-Monitor Set and supporting test equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 53, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

PPP-T-60	Tape, Packaging, Waterproof
PPP-T-76	Tape, Pressure Sensitive Adhesive, Paper (For Carton Sealing)
PPP-B-636	Boxes, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3643	Connector, Plug, Electrical Series HN, Type UG-59E/U
MIL-C-3650	Connector, Coaxial Radio Frequency, Series LC
MIL-N-18307	Nomenclature and Identification for Electronics, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-S-19500	Semiconductor Device, General Specification for
MIL-T-28800	Test Equipment for Use with Electrical and Electronic Equipment, General Specification for
MIL-M-38510	Microcircuit, General Specification for
MIL-C-38999	Connector, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded, and Breech Coupling), Environment Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for
MIL-C-39012	Connector, Coaxial, Radio Frequency, General Specification for
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities
MIL-C-55302	Connector, Printed Circuit Subassembly and Accessories
MIL-C-55544	Connector, Electrical, Environment Resistant for use with Flexible Flat Conductor Cable and Round Wire, General Specification for

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MIL-C-83723	Connector, Electrical (Circular, Environment Resistant), Receptacles and Plugs, General Specification for

STANDARDS

MILITARY

MIL-STD-108	Definition of Basic Equipment for Enclosure for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-147	Palletized Unit Loads Validated October 1974
MIL-STD-198	Capacitor, Selection and Use of
MIL-STD-199	Resistor, Selection and Use of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Requirements, Measurements of
MIL-STD-470	Maintainability Program Requirements for Systems and Equipments Prediction
MIL-STD-471	Maintainability Demonstrations
MIL-STD-701	List of Standard Semiconductor Devices
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests; Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Methods
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-1188	Commercial Packaging of Supplies and Equipments

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MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1562	Lists of Standard Microcircuits

HANDBOOKS

MILITARY

MIL-HDBK-472	Military Standardization Handbook, Maintainability Predictions

2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

1036AS3000	Distribution Box J-3425C/AWW

NAVAL AIR SYSTEMS COMMAND

NAVAIR 16-35J3425-1	Technical Manual, Intermediate Instruction with Illustrated Parts Breakdown Distribution Box J-3425C/AWW

BULLETINS

ANA Bulletin No. 400	Air Force-Navy Aeronautical Bulletin Electronic Equipment; Aircraft and Guided Missiles, Applicable Documents

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item description. The distribution box shall consist of an 11 x 11 x 4 inch metal case containing wires and terminals that interface with AN/AWW-12 or AN/AWW-9 Control-Monitor Set for test purposes.

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3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3 Materials, parts, and processes. Materials, parts, and processes used in the distribution box shall be as specified in MIL-T-28800, MIL-STD-454 and 3.3.1 through 3.3.2.2.

3.3.1 Parts. All parts shall be selected from Government furnished baseline or those parts included in ANA Bulletin No. 400, unless the parts listed therein will not perform as required in the intended applications and environments specified herein. If the latter is the case, request for use of nonstandard parts shall be obtained by the contractor in accordance with MIL-STD-965. Procedure I.

3.3.1.1 Solid state design. All microelectronic devices shall be selected as specified in Requirement 64 of MIL-STD-454 and MIL-STD-883. Hybrid microcircuits, including radio frequency, memory, microwave, and multimeter types are considered microelectronic devices and shall be controlled by Requirement 64 of MIL-STD-454. Hybrid and complex monolithic microcircuits are considered critical items and shall be treated as specified in MIL-STD-785. Any deviation from all solid state construction shall be authorized by the procuring activity prior to initiation.

3.3.1.2 Microcircuits and semiconductors. All microcircuits and semiconductors shall be packaged in hermetic packages. No plastic encapsulated devices shall be used.

3.3.1.2.1 Microcircuit devices. Only microcircuits listed in MIL-STD-1562 and procured as specified in MIL-M-38510 are standard for new design. When nonstandard microcircuit devices are approved for use as specified-in the general requirement of MIL-M-38510, Class B shall apply. Nonstandard devices shall be screened and qualified as specified in MIL-STD-883 (Test Methods 5004.3 and 5005.3). The qualification as specified in the requirements of Methods 5005.3, Group C, shall be satisfied by submission of approved test results for all standard and nonstandard microcircuit devices.

3.3.1.2.2 Semiconductor devices. Only JANTX semiconductor devices selected as specified in MIL-STD-701 are standard. When a JANTX device is not listed, the selection of nonstandard devices shall conform to the following order of precedences:

a. A JAN device listed as specified in MIL-STD-701.

b. A JAN device covered as specified by MIL-S-19500 but not listed in MIL-STD-701.

c. A commercial device.

The description of JANTX, JAN, and TX shall be specified by MIL-S-19500. As a minimum, a TX burn-in requirement as specified in Requirement 30 of MIL-STD-454 shall be required for all nonstandard devices. All semiconductors shall be selected as specified in Requirement 30 of MIL-STD-454. Only solid glass metallurgically bonded axial lead diodes and rectifiers shall be used. When TO-5 and TO-18 packages are required, they shall be limited to the solid metal header type. All semiconductor device junctions shall be protected and no organic or desiccant materials shall be

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included in the package. Thermocompression wedge bonding shall not be used with aluminum wire, Aluminum TO-3 packages shall not be used.

3.3. 1.2.2.1 Certification marks. The United States Government has adopted and is exercising legitimate control over the certification marks “JAN” and “J”, respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of military specifications. Accordingly, items acquired to, and meeting all of the criteria specified herein and in applicable specifications shall bear the certification mark “JAN” except that items too small to bear the certification mark “JAN” shall bear the letter “J”. The “JAN” or “J” shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the “JAN” or “J” may be located on the first line above or below the part number. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable specification shall not bear “JAN” or “J”. In the event an item fails to meet the requirements of this specification and the applicable specification sheets or associated detail specification, the manufacturer shall remove the “JAN” or “J” from the sample tested and also from all items represented by the sample. The “JAN” or “J” certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration No. 504,860 for the certification mark “JAN”.

3.3.1.3 Electrical connectors. Unless otherwise specified in the contract or purchase order (see 6.2.1), Requirement 10 of MIL-STD-454 shall be applicable except that only MIL-C-38999 and MIL-C-83723 covering circular connectors, and MIL-C-55302 covering printed circuit board connectors shall be employed. Connectors for use with flat cable shall meet the specified requirements of MIL-C-55544. RF coaxial connectors shall be as specified in MIL-C-39012, Series TNC, SC, SMA, or SMB, except where the power or voltage rating of SC series does not meet application requirements. In these cases, Series HN as specified in MIL-C-3643 or Series LC as specified in MIL-C-3650 shall be used.

3.3.1.4 Capacitors and resistors. Only established reliability capacitors and resistors as specified in MIL-STD-198 and MIL-STD-199 are standard.

3.3.2 Materials.

3.3.2.1 Moisture and fungus resistance. The distribution box shall be designed with materials that are moisture resistant and are not nutrient to fungi . The use of nutrient material is acceptable in hermetically sealed assemblies, paper capacitors, and transformers. Fungus inert materials are specified in MIL-STD-454, Requirement 4. Other materials shall be certified fungus-resistant or shall be tested as specified in 4.7.5. The parts used in the design shall be moisture resistant. Absorption, retention, or release of moisture by a part shall not degrade or affect performance.

3.3.2.2 Corrosion. Corrosion prevention and control as an overall concept shall be emphasized during design and fabrication. The distribution box shall be designed to prevent corrosion that would adversely affect its functions during the specified service life consistent with required reliability and maintenance concepts. Requirements 15 and 16 of MIL-STD-454 shall apply.

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3.3.3 Interchangeability. The distribution box shall meet the interchangeability requirements of MIL-T-28800. Drawing 1036AS3000 shall be standard for measuring interchangeability of all applicable units, assemblies, subassemblies, and parts.

3.4 Design and construction. The distribution box shall be manufactured in accordance with Drawing 1036AS3000, except that errors of omission or commission on the drawings may be incompatible with the requirements of this specification. In any case, it is a condition of final acceptance that the equipment manufactured shall meet all the design, construction, and workmanship requirements of this specification.

3.4.1 Reliability.

3.4.1.1 Operational stability. The distribution box shall operate with required performance, continuously or intermittently, for a period of not less than 1000 hours. During this time, it shall not be necessary to readjust any controls that are inaccessible to the operator during normal use.

3.4.1.2 Operating life. The distribution box shall have a total operating life of not less than 10,000 hours with minimal servicing and replacement of parts.

3.4.1.3 Reliability in mean-time-between-failures (MTBF). The distribution box shall have a specified mean (operating) time between failures of 800 hours when tested in accordance with 4.5.3.

3.4.2 Maintainability. Maintainability provisions shall be incorporated within the design to provide for service, access, and repair of all assemblies. The replacement of major components within the distribution box shall be accomplished without the need for special tools. Normal routine testing of the distribution box shall be accomplished using the standard test equipment normally available in intermediate maintenance shops.

3.4.2.1 Mean-time-to-repair (MTTR). The mean-time-to-repair (MTTR) of the distribution box shall be not greater than 2 hours when replacing circuit modules, plug-ins, assemblies, and parts that are intended for replacement at the intermediate level. Only where design changes shall cause significant deviations from procuring activity drawings, shall a prediction of MTTR be made in accordance with MIL-STD-470 and MIL-HDBK-472, Procedure II, and shall be demonstrated in accordance with 4.5.3.

3.4.3 Transportability. The distribution box shall be transportable by air, rail, truck, and ship when packaged as specified in section 5.

3.4.4 Standard conditions. The following conditions shall be used to establish the normal performance characteristics under standard conditions for performing laboratory bench tests:

a.	Temperature	Room ambient (30 ± 10°C).
b.	Altitude	Normal ground.
c.	Vibration	None.

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d.	Humidity	Room ambient up-to 90 percent relative humidity.
e.	Input Operating Voltage	115 ± 11.5 volts, 400 ± 20 Hz, 3Ø at 5 amperes, +28 +2 -6 volts dc at 11 amperes, +12 + .5 volts dc.

3.4.4.1 Service conditions. The distribution box shall operate as required under any of the environmental service conditions or combination of these conditions as specified in MIL-T-28800 for Class 3 equipment, As a minimum, the distribution box shall be required to meet specified performance while being exposed to the environmental conditions specified in 3.4.4.1.1 through 3.4.4.1.8.2.

3.4.4.1.1 High temperature. The distribution box shall operate as required during and after high-temperature exposure in accordance with 4.7.1.

3.4.4.1.2 Low temperature. The distribution box shall operate as required during and after low-temperature exposure in accordance with 4.7.2.

3.4.4.1.3 Humidity. The distribution box shall operate as required after being subjected to a humidity test in accordance with 4.7.3. The distribution box shall not be damaged as a result of the humidity test specified herein.

3.4.4.1.4 Watertight. The distribution box shall meet specified performance and accuracy requirements after being exposed to a watertight test in accordance with 4.7.4.

3,4.4.1.5 Fungus. The distribution box shall withstand, in both an operating and nonoperating condition, exposure of not less than 28 days in a fungi saturated atmosphere of ideal incubation conditions as specified in 4.7,5.

3.4.4.1.6 Altitude. The distribution box shall meet specified performance and accuracy requirements during and after being subjected to altitude test in accordance with 4.7.6.

3.4.4.1.7 Vibration. The distribution box shall meet the specified performance and accuracy requirements during and after being subjected to vibration test in accordance with 4.7.7.

3.4.4.1.8 Shock. The distribution box shall meet specified performance and accuracy requirements after being subjected to shock test in accordance with 4.7,8.

3.4.4.1.8.1 Transit drop. The distribution box shall meet specified performance and accuracy requirements after being subjected to drop test in accordance with 4.7.9.

3.4.4.1.8.2 Bench handling. The distribution box shall meet specified performance and accuracy requirements after being subjected to bench handling test in accordance with 4.7.10.

3.4.4.2 Warm-up time. The distribution box shall comply with specified performance requirements after a IO-minute warm-up period following a 2-hour nonoperating temperature stabilization period or as specified in the applicable test method.

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3.4.4.3 Input power. The distribution box shall be capable of specified operation when supplied input power that meets the requirements of MIL-T-28800. The power required shall be of the following types and shall be not greater than the following:

a. AC Power, 3-phase, 115 ± 11.5 volt, 400 ± 20 Hz, 5 amperes.

b. +28 +2, -6 Volts dc at 11 ampere capacity.

c. +12 ± .5 volts dc at 2 ampere capacity.

3.4.4.3.1 Protection. The distribution box shall be capable of withstanding input variance as imposed by abnormal electrical system operations as specified in MIL-STD-704.

3.4.4.4 Electromagnetic interference. The generation of electromagnetic interference by the distribution box and the susceptibility of the distribution box to electromagnetic interference shall be in accordance with MIL-STD-461 as referenced in MIL-T-28800.

3.4.4.4.1 Conducted emissions, power, and interconnecting leads. Conducted emissions from the distribution box power and interconnecting leads shall be within the limits of MIL-STD-461, Requirement CE03, over the frequency range of 0.015 to 50 MHz.

3.4.4.4.2 Conducted emissions, power leads, spikes, time domain. Conducted emission spikes and switching transients measured on the power leads in accordance with MIL-STD-462 shall be not greater than the limits of MIL-STD-461.

3.4.4.4.3 Conducted susceptibility, power leads. The distribution box shall not be susceptible to signals of 30 Hz to 400 MHz injected on the dc power leads at rms voltage levels of MIL-STD-461, Requirements CS01 and CS02.

3.4.4.4.4 Conducted susceptibility, spikes, power leads. The distribution box shall not be susceptible to spike interference injected on the dc power leads in accordance with MIL-STD-461, Requirement CS06.

3.4.4.4.5 Radiated emissions, electric fields, 14 KHz to 10 GHz. Radiated emission from the distribution box shall be within the limits of MIL-STD-461, Requirement RE02, over the frequency range of 14 KHz to 10 GHz.

3.4.4.4.6 Radiated susceptibility, magnetic induction field, spikes and power frequency. The distribution box shall not be susceptible to radiated fields in accordance with MIL-STD-461, Requirement RS02.

3.4.4.4.7 Radiated susceptibility, electric field, 14 KHz to 10 GHz. The distribution box shall not be susceptible to radiated fields in accordance with MIL-STD-461. Requirement RS03.

3.4.5 Dimensions. The dimensions of the distribution box shall be not greater than the following:

a. Length: 11 inches

b. Width: 11 inches

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c. Height: 4 inches

3.4.6 Weight. Weight of the distribution box shall be not greater than 25 pounds.

3.4.7 Color and finish. Color and finish of the distribution box shall be in accordance with Drawing 1036AS3000 and MIL-T-28800.

3.4.8 Nameplates and identification marking. Serial number assignment, nameplate approval, and identification marking shall be in accordance with MIL-N-18307.

3.4.8.1 Identification plates. Identification plates for the distribution box shall be in accordance with MIL-STD-130 and Drawing 1036AS3000.

3.4.8.2 Decals, nameplates, and marking locations. All decals, nameplates, and markings shall be affixed to the equipment in accordance with Drawing 1036AS3000 and all referenced marking specifications, documents, and drawings therein.

3.4.8.3 Serial numbers. Serial numbers will be assigned by the procuring activity for the equipment under contract or purchase order. The contractor shall serialize sequentially all equipments which are delivered under contract.

3.5 Performance. The performance of the distribution box shall be as specified herein, in addition to the performance requirements specified in Drawing 1036AS3000. Performance details are also provided in NAVAIR 16-35J3425C-1. Unless otherwise specifies herein, these requirements shall apply under both standard (see 3.4.4) and extreme service conditions (see 3.4.4.1). When reduced performance requirements for service conditions are acceptable, such variations are specified in the test requirement paragraph.

3.5.1 Function.

3.5.1.1 -12 Vdc binding post, E7. The E7 binding post shall provide an input connection for -12 volt power supply.

3.5,1.2 12 Vdc power ground binding post, El. The El binding post shall provide an input ground connection for 12 volt dc power supply.

3.5.1.3 +12 Vdc binding post, E2. The E2 binding post shall provide an input connection for +12 volt power supply.

3.5.1.4 CHANNEL SELECT rotary switch, S1. The S1 rotary switch shall be a nine-deck ten-position switch for selecting transmitter and receiver channels.

3.5.1.5 MSB, TP15. Test point, TP15, shall be provided for monitoring the Phase Data Link recognition code output of channel select BIT display board when testing XMTR power supply assembly.

3.5.1.6 MSB-1, TP16. Test point, TP16, shall be provided for monitoring the Phase Data Link recognition code output of channel select BIT display board when testing XMTR power supply assembly.

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3.5.1.7 ANT SELECT, TP21. Test point, TP21, shall be provided to check if aft antenna is selected.

3.5.1.8 MSB-2, TP17. Test point, TP17, shall be provided for monitoring the Phase Data Link recognition code output of channel select BIT display board when testing XMTR power supply assembly.

3.5.1.9 MSB, TP22. Test point, TP22, shall be provided for monitoring the output of the one wire channel select decoder.

3.5.1.10 MSB+2, TP18. Test point, TP18, shall be provided for monitoring the Phase Data Link recognition code output of channel select BIT display board when testing XMTR power supply assembly.

3.5.1.11 LSB+1, TP23. Test point, TP23, shall be provided for monitoring the output of the one wire channel select decoder.

3.5.1.12 MSB+, TP19. Test point, TP19, shall be provided for monitoring the Phase Data Link recognition code output of channel select BIT display board when testing XMTR power supply assembly.

3.5.1.13 LSB one wire, TP24. Test point, TP24, shall be provided for monitoring the output of the one wire channel select decoder.

3.5.1.14 LSB, TP20. Test point, TP20, shall be provided for monitoring the Phase Data Link recognition code output of channel select BIT display board when testing XMTR power supply assembly.

3.5.1.15 115V, 3Ø, 400 Hz toggle switch, CB1. Toggle switch, CB1, shall be a circuit breaker provided for turning power ON and OFF to the distribution box.

3.5.1.16 A XMTR, DS12; B XMTR, DS13; C XMTR, DS14; D XMTR, DS15; E XMTR, DS16. Light emitting diodes DS12 through DS16 shall be provided to indicate the transmitter channel selected in the transmitter power supply.

3.5.1.17 A RCVR, DS17; B RCVR, DS18; C RCVR, DS19; D RCVR, DS20; E RCVR, DS21. Light emitting diodes DS17 through DS21 shall be provided to indicate the receiver channel selected in the transmitter power supply.

3.5.1.18 CODE, S13. Switch, S13, shall be provided to set the recognition code for testing signal-control-processor (SCP).

3.5.1.19 1 WIRE/3 WIRE toggle switch, S11 and switch S14. Toggle switch, S11, and switch S14 shall be provided to select either one wire or three wire operation.

3.5.1.20 RECORDER REMOTE toggle switch, S2. Toggle switch, S2, shall be provided to control +28 volts to video tape recorder.

3.5.1.21 TRIGGER toggle switch, S3. Toggle switch, S3, shall be provided for an ON/OFF switch to control +28 volts to the AN/AWW-12 or AN/AWW-9 Control-Monitor Set or the SCP.

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3,5.1.22 STATION SELECT toggle switch, S4. Toggle switch, S4, shall be provided for an ON/OFF switch to control +28 volts to the AN/AWW-12 or AN/AWW-9 Control-Monitor Set to enable the control monitor set or SCP to accept a multiplex signal.

3.5.1.23 PSI CONTROL toggle switch, S5. Toggle switch, S5, shall be provided for an ON/OFF switch to control +28 volts to the AN/AWW-12 or AN/AWW-9 Control-Monitor Set or SCP to activate ARM logic and 2.3 KHz oscillator in the SCP.

3.5.1.24 HI/POWER toggle switch, S6. Toggle switch, S6, shall be provided to apply ground to transmitter, in the 1 A/C mode, for attenuator relay.

3.5.1.25 XMTR toggle switch, S7. Toggle switch, S7, shall be provided for an ON/OFF switch to connect transmitter power supply VR1 to ground.

3.5.1.26 ANT SELECT toggle switch, S8. Toggle switch, S8, shall be provided for an ON/OFF switch to control antenna select from forward to aft antenna.

3.5.1.27 BIT AFT ANT/BIT FWD ANT toggle switch, S9. Toggle switch, S9, shall be provided to select the forward or aft antenna from the distribution box.

3.5.1.28 BIT toggle switch, S10. Toggle switch, S10, shall be provided for an ON/OFF switch to control +28 volts to transmitter local oscillator for RF loop built-in test.

3.5.1.29 1 A/C/2 A/C toggle switch, S12. Toggle switch, S12, shall be provided to complete the circuit in SCP for one aircraft operation.

3.5.1.30 LP ENABLE, DS11. Light emitting diode, DS11, shall be provided to indicate that 1 A/C/2 A/C switch in AN/AWW-12 or AN/AWW-9 Control-Monitor Set is in the 1 A/C position.

3.5.1.31 XMTR POWER, DS9. Light emitting diode, DS9, shall be provided to indicate that the transmitter is providing RF.

3.5.1.32 STATION SELECT, DS8. Light emitting diode, DS8, shall be provided to indicate that SCP has acquired an internal path for STATION SELECT signal from cockpit to the transmitter.

3.5.1.33 HI POWER, DS7. Light emitting diode, DS7, shall be provided to indicate that the SCP is supplying a ground to 30 dB attenuator relay in the transmitter.

3.5.1.34 ANT SELECT, DS6. Light emitting diode, DS6, shall be provided to indicate that SCP is supplied with a ground signal for the ANT SELECT relay in the transmitter.

3.5.1.35 XMTR ON, DS5. Light emitting diode, DS5, shall be provided to indicate that ground is applied to transmitter power supply VR1.

3.5.1.36 BIT AFT ANT, DS4. Light emitting diode, DS4, shall be provided to indicate that ground is being applied to transmitter AFT test relays.

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3.5.1.37 BIT FWD ANT, DS3. Light emitting diode, DS3, shall be provided to indicate that ground is being applied to the FWD test relay in the transmitter.

3.5.1.38 PCO, DS2. Light emitting diode, DS2, shall be provided to indicate a PCO signal has been sent by the AN/AWW-12 or AN/AWW-9 Control-Monitor Set.

3.5.1.39 VTR REMOTE, DS1. Light emitting diode, DS1, shall be provided to indicate the presence of +12 volts from the VTR REMOTE switch in the SCP.

3.5.1.40 VCO, DS10. Light emitting diode, DS10, shall be provided to indicate that the VCO is generating RF.

3.5.1.41 -28 Vdc binding post, E3. Binding post, E3, shall provide an input ground connection for the -28 Vdc.

3.5.1.42 +28 Vdc binding post, E4. Binding post, E4, shall provide an input connection for the +28 Vdc.

3.5.1.43 VTR VIDEO IN, J2. Connector, J2, shall provide an input connection for the video generator.

3.5.1.44 Connector, J13. Connector, J13, shall connect distribution box to the receiver, video tape recorder, and the one wire demodulator.

3.5.1.45 VTR VIDEO OUT, J3. Connector, J3, shall be provided for connecting the distribution box to the video monitor.

3.5.1.46 VTR VIDEO IN, J4. Connector, J4, shall be provided for connecting AUDIO to the video tape recorder.

3.5.1.47 VTR AUDIO OUT, J5. Connector, J5, shall be provided for monitoring the audio signal being recorded on the video tape recorder.

3.5.1.48 Connector, J14. Connector, J14, shall be provided for connecting the distribution box to the SCP.

3.5.1.49 EP VIDEO OUT, J6. Connector, J6, shall be provided for monitoring video signal from the SCP to the video tape recorder.

3.5.1.50 EP AUDIO IN, J7. Connector, J7, shall be provided to apply audio modulation to the video signal via the SCP.

3.5.1.51 EP AUDIO OUT, J8. Connector, J8, shall be provided for monitoring the audio signal the from the SCP.

3.5.1.52 Connector, J15. Connector, J15, shall be provided to connect the distribution box to the transmitter.

3.5 1.53 MULTIPLEX, J9. Connector, J9, shall be provided to apply multiplex signals to the SCP.

3.5.1.54 MOD OUT, J10. Connector, J10, shall be provided for monitoring the modulation output of the SCP.

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3.5.1.55 COCKPIT VIDEO, J11. Connector, J11, shall be provided for monitoring the video output of the SCP.

3.5.1.56 XMTR MOD, J12. Connector, J12, shall be provided for applying external modulation to the transmitter.

3.5.1.57 Connector, J16. Connector, J16, shall be provided for connecting the distribution box to the transmitter power supply.

3.5.1.58 GRD, TP14. Test point, TP14, shall be provided as a common ground for all test points.

3.5.1.59 NEUTRAL, TP13. Test point, TP13, shall be provided as a common ground for ac test points.

3.5.1.60 ØC, TP12. Test point, TP12, shall be provided for checking transmitter 115 Vac ØC from the SCP.

3.5.1.61 ØB, TP11. Test point, TP11, shall be provided for checking transmitter 115 Vac ØB from the SCP.

3.5.1.62 ØA, TP10. Test point, TP10, shall be provided for checking transmitter 115 Vac ØA from the SCP.

3.5.1.63 ØC, TP9. Test point, TP9, shall be provided for checking transmitter 115 Vac ØC from the SCP.

3.5.1.64 ØB, TP8. Test point, TP8, shall be provided for checking transmitter 115 Vac ØB from the SCP.

3.5.1.65 ØA, TP7. Test point, TP7, shall be provided for checking transmitter 115 Vac ØA from the SCP.

3.5.1.66 BIT +28V, TP6. Test point, TP6, shall be provided for monitoring +28 Vdc during bit testing.

3.5.1.67 -12V, TP5. Test point, TP5, shall be provided for monitoring -12 Vdc from the SCP.

3.5.1.68 -6V, TP4. Test point, TP4, shall be provided for monitoring -6 Vdc from the SCP.

3.5.1.69 +5V, TP3. Test point, TP3, shall be provided for monitoring +5 Vdc from the SCP.

3.5.1.70 +12V, TP2. Test point, TP2, shall be provided for monitoring +12 Vdc from the SCP.

3.5.1.71 +28V, TP1. Test point, TP1, shall be provided for monitoring +28 Vdc from the SCP.

3.5.1.72 Circuit board Al, circuit board A2, and oscillator Y1. Circuit board Al, circuit board A2, and oscillator Y1, shall provide the capability to test the one wire channel select function of the SCP or the one wire demodulator in the AN/AWW-12 or AN/AWW-9 Control-Monitor Set.

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3.6 Detail requirements.

3.6.1 Front panel assembly. The front panel assembly shall contain all the controls, indicators, and power supply input terminals. The front panel assembly shall be manufactured in accordance with Drawing 1036AS2018.

3.6.2 Case assembly. The case assembly shall consist of the case on which shall be mounted connectors, conduit box connector, conduit coupling locknut, cable strap mount, and rubber feet. The case assembly shall be manufactured in accordance with Drawing 1036AS2021.

3.6.3 Card holder assembly. The card holder assembly shall contain two circuit card assemblies. The card holder shall be manufactured in accordance with Drawing 1036AS510. The two circuit card assemblies shall be manufactured in accordance with Drawings 1036AS501 and 1036AS505.

3.6.4 Terminal board assembly. The terminal board assembly shall be manufactured in accordance with Drawing 1036AS2012.

3.6.5 Case, cable storage. The cable storage case shall be manufactured in accordance with Drawing 1036AS2015.

3.6.6 Power cable assembly, W1. The W1 power cable assembly shall connect the distribution box to 115V, 400 Hz 3Ø power source which in turn distributes the power to the unit under test. The power cable assembly shall be manufactured in accordance with Drawing 1036AS112.

3.6.7 Cable, W2. Cable, W2, shall connect the distribution box to the AN/AWW-12 or AN/AWW-9 Control-Monitor Set when checking the Control-monitor Set as a system.

3.6.8 Cable assembly, W3, receiver-VTR AN/AWW-12. The W3 cable assembly shall connect operating voltage and signal input/output between the distribution box and the video tape recorder, supply three-phase power and channel selection to the receiver. The W3 cable assembly shall be manufactured in accordance with Drawing 1036AS2003.

3.6.9 Cable assembly, W4, signal control processor AN/AWW-12. The W4 cable assembly shall connect operating voltages and signal input/output between the distribution box and the AN/AWW-12 Signal Control Processor. The W4 cable assembly shall be manufactured in accordance with Drawing 1036AS2004.

3.6.10 Cable assembly, W5, transmitter AN/AWW-12. The W5 cable assembly shall connect operating voltages and signal input/output between the distribution box and the AN/AWW-12 Transmitter. The W5 cable assembly shall be manufactured in accordance with Drawing 1036AS2005.

3.6.11 Cable assembly, W6, receiver-VCTR, AN/AWW-9. The W6 cable assembly shall connect operating voltages and signal input/output to the video cassette tape recorder and shall supply the one-wire demodulator, channel selection, and three-phase power to the receiver. The W6 cable assembly shall be manufactured in accordance with Drawing 1036AS2006.

3.6.12 Cable assembly, W7, signal control processor, AN/AWW-9. The W7 cable assembly shall connect operating voltages and signal input/output between the distribution box and the AN/AWW-9 Signal Control Processor. The W7 cable assembly shall be manufactured in accordance with Drawing 1036AS2007.

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3.6.13 Cable assembly, W8, transmitter, AN/AWW-9. The W8 cable assembly shall connect operating voltages and signal input/output between the distribution box and the AN/AWW-9 Transmitter. The W8 cable assembly shall be manufactured in accordance with Drawing 1036AS2008.

3.6.14 Cable assembly, W9, transmitter power supply, AN/AWW-9. The W9 cable assembly shall connect operating voltages and signal input/output between the distribution box and the AN/AWW-9 Transmitter Power Supply. The W9 cable assembly shall be manufactured in accordance with Drawing 1036AS2009.

3.7 Safety. The contractor shall ensure that safety requirements are in accordance with MIL-T-28800.

3.7.1 Human performance/human engineering. The distribution box and all component parts thereof, shall be designed in accordance with MIL-STD-1472. Recognized principles of human engineering as specified in MIL-H-46855 shall be followed during design.

3.8 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9 and Drawing 1036AS3000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sample tests of quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the conditions specified in 3.4.4.

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4.4 First article inspection. First article inspection shall be made on one unit representative of the production equipments to be supplied under the contract. First article inspection shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedures (see 4.11). Column I of table I lists minimum tests required for first article inspection. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government. No first article inspections shall be conducted prior to approval of first article test procedure by the procuring activity.

4.4.1 Scope of first article tests. first article tests shall include all tests listed in table I, column I and any additional tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of-this specification.

4.4.2 First article approval. Approval of first article sample shall be by the procuring activity upon satisfactory completion of all tests.

4,5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance inspection. All inspection and testing shall be under the supervision of the Government inspector. Contractors-not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. Quality conformance inspection shall-consist of the following tests:

a. Individual tests (see 4.5.1).

b. Sample tests (see 4.5.2).

c. Special tests (see 4.5.5).

d. Life tests (see 4.6).

4.5.1 Individual tests. Each distribution box submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine-compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each distribution box accepted shall have-passed the following tests:

a. Examination of product (see 4.5.1.1).

b. Operational test (see 4.5.1.2).

c. Manufacturing run-in test (see 4.5.1.3).

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TABLE I. Test requirements

Requirement	I. First Article Inspection	II. Quality Conformance Inspections			
		Individual (A)	Sampling (B)	Special (C) 1/	Life (D)
Visual Examination (3.3, 3.4, 3.4.5 through 3.4.8.3, and 3.8)	4.5.1.1	4.5.1.1	4.5.1.1	4.5.1.1	4.5.1.1
Performance Characteristics (3.5)	4.5.1.2	4.5.1.2	4.5.1.2	4.5.1.2	4.5.1.2
Workmanship (3.8)	4.5.1.3	4.5.1.3	4.5.1.3	4.5.1.3	4.5.1.3
Environmental (3.4.4.1)	4.7				(See 4.7.1, 4.7.2
Warm-up time (3.4.4.2)	4.8				
Power protection (3.4.4.3.1)		1/	2/	3/	and 4.7.3)
Interference Control (3.4.4.4)	4.10				
Reliability (3.4.1)	4.5.3		4.5.3.2		
Maintainability (3.4.2)	4.5.4				

1/ Additional individual tests may be requested by the procuring activity (see 4.5.1 and 6.2.1).

2/ Sampling tests to be performed will be determined by the procuring activity based on first article and individual test data (see 4.5.2.1).

3/ Special tests to be performed will be determined by the procuring activity (see 4.5.5.2).

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4.5.1.1 Examination of product. Each distribution box shall be examined carefully to determine that the material and workmanship requirements have been met. Attention shall be given, but not limited to the following:

a. Completeness.

b. Nameplates, identification markings and labels.

c. Finishes.

d. Welded and soldered joints.

e. Fit of components.

f. Mounting of components.

g. Safety features and interlocks.

h. Fastening and securing of devices and parts.

i. Cables including plugs and receptacles.

j. Grounding connections.

k. Weight.

l. Overall dimensions.

m. Potential shorts and defective insulation.

n. Foreign objects, lint, dust.

o. Other visual defects.

4.5.1.2 Operational test. Each distribution box shall be operated long enough to permit the distribution box temperature to stabilize and to check sufficient characteristics and record adequate data to assure the distribution box operation meets the requirements of this specification.

4.5.1.3 Manufacturing run-in test. Each distribution box shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be specified as anything which causes malfunctioning of the equipment.

a.	Temperature	Ambient room
b.	Humidity	Ambient room
c.	Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ±3g's

The distribution box shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the distribution box shall be operated during this vibration for the purpose of detecting flaws and imperfect workmanship.

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The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the distribution box in two directions the vertical direction shall be used. During the 6-hour period of operations following the 10-minute vibration period, the distribution box shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it shall be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration.

4.5.2 Sampling tests. Distribution boxes selected for sampling tests shall first have passed the individual tests. Distribution boxes will be selected for sampling tests by the Government inspector in accordance with table II.

TABLE II. Sampling test sample selection

Quantity of Distribution Boxes Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.5.2.1 Scope of sampling tests. As a minimum, each distribution box selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each distribution box tested and should be based on results of the first article, individual, and special tests.

c. Manufacturing run-in test specified in 4.5.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures.

4.5.3 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with MIL-STD-781. Tests as required by both the qualification phase and the production acceptance (sampling) phase shall be conducted.

4.5.3.1 Qualification phase. Prior to the acceptance of distribution boxes under the contract or purchase order, a minimum of three distribution boxes shall be tested in accordance with MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of distribution boxes to be used shall be those listed in table V of MIL-STD-781. For the qualification phase, Test Level E shall be used. The accept-reject criteria for Test Plan IV shall be used.

4.5.3.2 Reliability sampling phase tests. Samples of the distribution box shall be tested in accordance with MIL-STD-781 under the section entitled

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"Sampling Phase of Production Reliability Tests." For the sampling phase, Test Level E shall be used, except that the distribution box shall operate continuously. The accept-reject criteria for Test Plan V shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.5.3.2.1 Lot size for sampling phase. The distribution boxes constructed during a 2-month period shall be one lot, The accept-reject criteria for Test Plan IV shall be used to determine the length of the tests in each lot. Unless otherwise specified in the contract or purchase order, no distribution boxes shall be shipped until an accept decision is reached under each lot (see 6.2.1).

4.5.3.3 Test details. The test details shall be part of the test procedures prepared prior to the beginning of the qualification test phase of the reliability assurance test.

4.5.4 Maintainability test. The Mean-Corrective-Maintenance-Time and Maximum-Corrective-Maintenance-Time as specified in 3.4.2 shall be demonstrated as specified in MIL-STD-471. Test Method 9.

4.5.5 Special tests. Special tests shall be conducted on a quantity of distribution boxes for the purpose of checking the effect of any design or material change on the performance of the distribution box and to assure adequate quality control. The distribution box selected for special tests may be selected from distribution boxes previously subjected to the sampling or reliability assurance tests.

4.5.5.1 Special test schedule. Selection of distribution boxes for special tests shall be made as follows:

a. On an early distribution box, after an engineering or material change.

b. Whenever failure reports or other evidences indicate additional tests are required. (This will be determined by the procuring activity.)

4.5.5.2 Scope of special tests. Special tests shall consist of such tests acceptable to the procuring activity. Test procedures previously approved for the first article tests shall be used where applicable.

4.5.6 Distribution box failure. Should a failure occur during either the sample or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

c. Where practical, include a test (in the individual test) to check all distribution boxes for this requirement until assurance is obtained that the defect has been corrected.

4.6 Life test. The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on distribution boxes that have passed the individual tests. The life test shall be performed under the conditions

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specified in 4.6.1. The life test sample shall be selected by the Government inspector in accordance with table III. (Distribution boxes which have successfully passed the Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of distribution boxes equal to, or more than, that listed in table III receive at least 300 hours each of test time.

TABLE III. Life test sample selection

Quantity of Distribution Boxes Offered for Acceptance	Quantity to be Selected for Sampling Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.6.1 Test conditions. The life test shall be conducted under the standard conditions specified in 3.4.4.

4.6.2 Test periods. The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the distribution box temperature to stabilize. Periodically, the distribution box shall be turned on and off several times and put through its various phases of operation

4.6.3 Performance check. At approximately 8-hour intervals during the test, a limited performance check shall be made.

4.6.4 Test data. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period.

4.6.4.1 Failure report. In the event of a failure, the Government inspector shall be notified immediately.

4.7 Environmental tests. The distribution box shall be subjected to environmental tests specified in 4.7.1 through 4.7.8.

4.7.1 High temperature. The distribution box shall be tested in accordance with MIL-STD-810, Method 501.2, Procedure I modified as follows

a. Step 1. Perform operation tests of 4.5.1.2 under the closely controlled conditions of MIL-STD-810.

b. Step 2. The high storage temperature shall be 85°C. The storage temperature shall be maintained for 24 hours. The storage temperature shall be 85° for 4 hours and 75°C for the following 20 hours.

c. Step 4. The highest continuous operating temperature shall be 55°C. The distribution box shall be operated for 20-minutes intermittently at 71°C after the required warm-up time. The relative humidity shall be between 20 and 35 percent.

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d. Step 5. Perform operation tests of 4.5.1.2.

e. Step 6. The distribution box shall be operated during the return to standard ambient conditions.

f. The rate of temperature change (steps 2, 4, and 6) shall be between 1° and 5°C per minute.

4.7.2 Low temperature. The distribution box shall be subjected to low temperature tests in accordance with MIL-STD-810, Method 502.2, Procedure I modified as follows.

a. Step 1. Perform the operation tests of 4.5.1.2 under the closely controlled conditions of MIL-STD-810.

b. Step 2. The low storage temperature shall be -62°C. The relative humidity shall be uncontrolled during temperature transition and while at storage temperature. The storage temperature shall be maintained for 24 hours, The distribution box shall be maintained at -62°C for 4 hours and then -55°C for the following 20 hours.

c. Step 4. The lowest operating temperature shall be -40°C. Stabilize the test set temperature for 2 hours.

d. Step 5. Energize the distribution box and allow the warm-up time specified in 3.4.4.2.

e. Step 6. The distribution box shall be operated during the return to controlled ambient conditions. Stabilize the test set temperatures for 2 hours.

f. Step 7. Perform the operation test of 4.5.1.2.

g. The rate of temperature change (steps 2, 4, and 6) shall be between 1° and 5°C per minute.

4.7.3 Humidity. The distribution box shall be subjected to the humidity test specified in MIL-STD-810, Method 507.2, Procedure III.

4.7.4 Watertight. The distribution box shall be watertight tested in accordance with MIL-STD-108. The distribution box shall be operated during an after exposure.

4.7.5 Fungus. Susceptibility of the distribution box to fungus growth shall be determined by examining the list of materials using the criteria in MIL-STD-454, Requirement 4, to identify fungi nutrient materials.

4.7.6 Altitude. The distribution box shall be subjected to an altitude test in accordance with MIL-STD-810, Method 500.2, Procedure I, except the simulated altitudes shall be 15,000 feet (operating) and 40,000 feet (nonoperating). The following details shall modify the MIL-STD-810 when necessary:

a. Pretest data: Perform the operation test of 4.5.1.2.

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b. Length of time required for operation and measurements: warm-up or stabilization time as required in 3.4.4.2, plus a minimum of 2 hours operation in the operation test of 4.5.1.2.

c. Perform the operation test after return to standard ambient conditions. Degradation of distribution box performance beyond the specified requirements of 4.5.1.2 shall constitute a failure,

4.7.7 Vibration. The distribution box shall be subjected to a vibration test in accordance with MIL-STD-810, Method 514.3, Procedure X, modified as follows:

a. The distribution box shall be attached directly to the vibration table or to an intermediate structure which is so designed as to be capable of transmitting the specified magnitudes of vibration to the points of the distribution box attachment throughout the required test frequency range.

b. During vibration the distribution box shall be operating in the specified operation check mode.

c. Perform the operation tests required in 4.5.1.2.

4.7.8 Shock. The distribution box shall be subjected to a shock test in accordance with MIL-STD-810. Method 516.3, Procedure I, using the half-sine shock waveform of the Half Sine Shock Pulse Configuration and its Tolerance Limits figure of FIIL-STD-810, nominal duration (D) of 11 ms. The distribution box shall be given three shocks in each direction and each axis (total 18). Peak value (A) of each shock shall be 30 g’s. The distribution box shall be operated during the shock. After the shock exposure, perform the operation test specified in 4.5.1.2.

4.7.9 Transit drop. The distribution box shall be subjected to a transit drop test in accordance with MIL-STD-810. Method 516.3, Procedure II, using a drop height of 760mm (30 inches). The distribution box shall be dropped once on each corner and each face for a total of 14 drops. Perform the operation test specified in 4.5.1.2.

4.7.10 Bench handling. The distribution box shall be subjected to the bench handling test of MIL-STD-810, Method 516.3, Procedure V, modified as follows:

a. The distribution box shall be tested while operating in the operation test of 4.5.1.2.

b. Examine the distribution box for mechanical damage. Damage to any control , indicator, fuseholder, connector, or other protruding component part shall constitute a failure.

c. Perform the operation test of 4.5.1.2.

4.11 Warm-up time. The distribution box shall be subjected to tests to determine the capability of the distribution box to conform to the warm-up time requirements of 3.4.4.2.

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4.12 Input power test.

4.12.1 Voltage and frequency variation tests. The distribution box response to the input voltage and frequency variations shall be tested in accordance with MIL-T-28800. Operational performance may be degraded during the exposure but distribution box shall not sustain damage and shall return to normal operation when test is completed,

4.12.2 Voltage and frequency-transient test. The distribution box performance during voltage transients shall be tested in accordance with MIL-T-28800. Operational performance may be degraded during the exposure but distribution box shall not sustain damage and shall return to normal operation when test is completed.

4.13 Electromagnetic interference (EMI) tests. EMI tests shall be performed in accordance with MIL-STD-462 to determine that the distribution box meets the requirements of MIL-STD-461 for Class B equipment (see 3.4.4.4), The distribution box shall be operating during the EMI tests. The distribution box shall be monitored for signs of vulnerability during susceptibility tests.

4.13.1 Conducted emissions, power and interconnecting leads, 0.015 to 50 MHz. The distribution box shall be subjected to a conducted emission test on the power and interconnecting leads in accordance with MIL-STD-462, Method CE03.

4.13.2 Conducted emissions, power leads, spikes, time domain. The distribution box shall be subjected to a conducted emission test on the power leads in accordance with MIL-STD-462, Method CE07.

4.13.3 Conducted susceptibility power leads, 30 Hz to 50 KHz. The distribution box shall be subjected to signals between 30 Hz to 50 KHz being injected on the dc power leads in accordance with MIL-STD-462, Method CS01.

4.13.4 Conducted susceptibility power leads, 0.05 to 400 MHz. The distribution box shall be subjected to signals between 0.05 to 400 MHz being injected on the dc power leads in accordance with MIL-STD-462, Method CS02.

4.13.5 Conducted susceptibility, spikes, power leads. The distribution box shall be subjected to the injection of spikes on all ungrounded ac and dc input power leads in accordance with MIL-STD-462, Method CS06.

4.13.6 Radiated emissions, electric field, 14 KHz to 10 GHz. The distribution box shall be probed for points of maximum radiation over the frequency range of 14 KHz to 10 GHz in accordance with MIL-STD-462, Method RE02.

4.13.7 Radiated susceptibility, magnetic induction field, spikes and power frequency. The distribution box shall be subjected to test spikes each having a waveform in accordance with MIL-STD-462, Method RS02, Part I and subjected to 20 amperes applied to a test wire at power frequencies of the distribution box in accordance with MIL-STD-462, Test Method RS02, Part II.

4.13.8 Radiated susceptibility, electric field, 14 KHz to 10 GHz. The distribution box shall be subjected to an electric field in the frequency range of 14 KHz to 10 GHz in accordance with MIL-STD-462, Method RS03.

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4.14 Test procedures. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract.

4.15 Presubmission testing. No item, part, or complete distribution box shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements of this specification and the contract.

4.16 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of Section 5 and the standards and specifications listed therein.

5. PACKAGING

5.1 Preservation. Preservation shall be level C or industrial as specified (see 6.2.1).

5.1.1 Level C.

5.1.1.1 Cleaning. Distribution boxes shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Distribution boxes shall be dried in accordance with MIL-P-116.

5.1.1.3 Unit packs. Unless otherwise specified in the contract, each distribution box shall be individually unit packed in accordance with the methods of MIL-P-116 designated herein ensuring compliance with the applicable requirements of that specification (see 6.2.1).

5.1.2 Industrial. The Industrial preservation of distribution boxes shall be in accordance with the requirements of MIL-STD-1188.

5.2 Packing. Packing shall be level C or industrial, as specified in the contract (see 6.2.1).

5.2.1 Level C. The packaged distribution boxes shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the Appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer’s joints with tape, 2-inches minimum width, conforming to PPP-T-60, Class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the Appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Industrial. The preserved distribution boxes shall be packed in accordance with the requirements of MIL-STD-1188.

5.2.3 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or purchase order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

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5.2.3.1 Level C. Distribution boxes, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.3 Marking.

5.3.1 Level C. In addition to any special or other identification marking specified in the contract (see 6.2.1), each unit pack, intermediate and exterior container, and unitized load shall be marked in accordance with MIL-STD-129.

5.3.2 Industrial. Industrial marking shall be in accordance with the requirements of MIL-STD-1188.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1 and 5.2.2) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.16.

6. NOTES

6.1 Intended use. The Distribution Box J-3425C/AWW is intended for use at Intermediate Level shops to interface with support equipment to test AN/AWW-12 or AN/AWW-9 Control-Monitor Set.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of packaging an packing (see 5.1).

c. When a first article is required for inspection and approval (see 4.4).

d. Additional individual tests, if required (se 4.5.1 and table II).

e. Lot acceptance for each lot, if other than specified herein (see 4.5.3.2.1).

f. Unit packs, if other than specified herein (see 5.1.1.3).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DO Form 1423 is not used, the data specified below shall be delivered by the

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contractor in accordance with the contract or purchase order requirements, Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Options
4.4.4.11	Procedure, First Article Inspection	DI-T-4901	---
4.4.1, 4.4.3	Report, First Article Inspection	DI-T-4902	---
4.5.5, 4.5.5.2, 4.6.3, 4.11	Procedures, Production/Acceptance Inspection	DI-T-4903	---
4.5, 4.6.4	Reports, Production Inspection	DI-T-4904	---
4.5.3.3	Procedures, Reliability Tests	DI-R-7035	---
4.6.4.1	Engineering Change Proposals (ECPs) and Request for Deviations and Waivers (Long Form)	DI-E-2037	---
4.5.6c	Plan, Failure Data Collection, Analysis and Corrective Action	UDI-T-23719	---
4.6.4.1	Report, Failure	DI-R-21598	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5010.12-L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the distribution box will be tested and should be a first article sample. The first article should consist of one distribution box. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (keyword) listing.

Antenna
Control-monitor set
Distribution box
One wire channel select decoder

Preparing Activity
Navy-AS
(Project 4935-N065)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-85859(AS)	DISTRIBUTION BOX J-3425C/AWW, GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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